Cat Crusher

Short manual . free and open source (AGPL-3.0-or-later) . github.com/SpeakingColors

Cat Crusher is a distortion, filter and compressor in one plugin. It makes drums, bass and leads thicker, grittier and louder: choose one of ten distortion types, add a hard digital crunch with Mech, shape the tone with a resonant 24 dB filter (low-pass, high-pass or band-pass), then glue everything with a one-knob compressor with automatic make-up gain, so the level stays the same at any Amount and only the density changes. A hidden soft clipper and safety limiter keep peaks under control. Built for dance, house, UK hardcore and hardstyle.

Quick start

• Pick the preset closest to your sound (Fat Bass, Hardstyle Kick Crunch, UK Hardcore Stab, Gritty Drums).
• Set Type first, then raise Drive until it sounds right; the Distortion curve shows the shape you are using.
• Use Cutoff to remove harsh highs (Filter type Low-pass), thin the lows (High-pass) or make a telephone or radio sound (Band-pass), Amount for density, and Mix for parallel crush.
• Compare with the plugin bypassed and set Volume so both are equally loud.

How the window works

• Each knob sits directly under the display it changes; the displays react to the knobs and the knobs react to the displays.
• Meters show the real signal in dB with a dB scale: input and output on the far right.
• Hover a control for about one second to see an info bubble. Turn bubbles off with the round i button in the header.
• The Limiter switch in the header adds a safety limiter at the output (useful while experimenting).
• Presets are in the header; the dice picks random settings (some controls are excluded on purpose).

Controls

Distortion

Distortion	Choice: Off / On (default On)	Distortion on/off.
Type	Choice: Tube, Soft Sat, Transistor, Tape, Diode, Hard Clip, Fuzz, RAT, Overdrive, Rectifier (default Tube)	Distortion type. Tube: warm even harmonics. Soft Sat: smooth and symmetric, great on buses. Transistor: console-style, punchy and forward. Tape: gentle and rounded. Diode: edgy and asymmetric. Hard Clip: tight, loud and aggressive. Fuzz: thick, gated and woolly. RAT: hard-clipped, gritty distortion pedal style. Overdrive: smooth, mid-forward tube-pedal style. Rectifier: adds a bright octave-ish edge.
Drive	0 to 100 % default 30 %	Drive: how hard the selected distortion type is pushed. Higher = more saturation and harmonics.
Mech	0 to 100 % default 0 %	Mech: harder, more aggressive digital-style crunch (folds and clips the wave). 0 = off.

Filter

Filter type	Choice: Low-pass, High-pass, Band-pass (default Low-pass)	Filter type: Low-pass keeps the lows and cuts the highs, High-pass keeps the highs and cuts the lows, Band-pass keeps only a band around the cutoff.
Filter	Choice: Off / On (default On)	Filter on/off.
Cutoff	40 to 20000 Hz default 20000 Hz	Cutoff: frequency of the 24 dB filter (low-pass, high-pass or band-pass). As a low-pass it rolls off the harsh highs that distortion creates.
Res	0 to 100 % default 10 %	Resonance: boosts the frequencies right at the cutoff for a vocal or squelchy edge.

Compressor

Compressor	Choice: Off / On (default On)	Compressor on/off.
Phat mode	Choice: Off / On (default Off)	Phat mode: adds low-end weight before the compressor, a slower release and a soft ceiling. Great on kicks and bass.
Amount	0 to 100 % default 30 %	Compressor amount: lowers the threshold and raises the ratio together. Make-up gain is automatic, so the level stays the same at any amount and only the density changes.

Master

Volume	-24 to 12 dB default 0 dB	Volume: final output level. Match it to the bypassed level when comparing.
Mix	0 to 100 % default 100 %	Mix: dry/wet blend. Lower values keep the original punch (parallel crush).

Recommendations

- Kick: Type Diode or Transistor, Drive 60 to 80, Phat mode on, Mix 80 to 90 to keep the click.
- Bass: Type Transistor, Drive 50, Cutoff 4 kHz, Phat mode on; keep Mech low so the sub stays clean.
- Radio or telephone: Filter type Band-pass, Cutoff 1 to 3 kHz, Resonance 30 to 60, Mech 40.
- Lead or stab: Mech 30 to 60 and Resonance 30 to 50 for screech; roll Cutoff down if it gets harsh.
- Drum bus: Type Tape, Drive 20 to 30, Amount 25 to 40, Mix 100 for glue or 60 to 70 for parallel punch.
- If it sounds louder but worse, lower Volume: loudness fools the ear.

Presets

Init, Warm Bus, Gritty Drums, Fat Bass, Radio Crush, Screech Lead, Hardstyle Kick Crunch, UK Hardcore Stab, Smooth Lows, Telephone Crunch, Mech Mayhem

Install and use

- Windows: run the installer (.exe). macOS: open the installer (.pkg or .dmg). Linux: install the Flatpak with `flatpak install --user` and the .flatpak file, or copy the .vst3 folder to ~/.vst3.
- Everything works offline. Rescan plugins in your DAW, then look for the plugin under Studio 87.
- Report problems or ideas on the GitHub page of the plugin. Free to use, copy and change under the AGPL-3.0-or-later licence.